

Hand-on Training on GIS Development

By:

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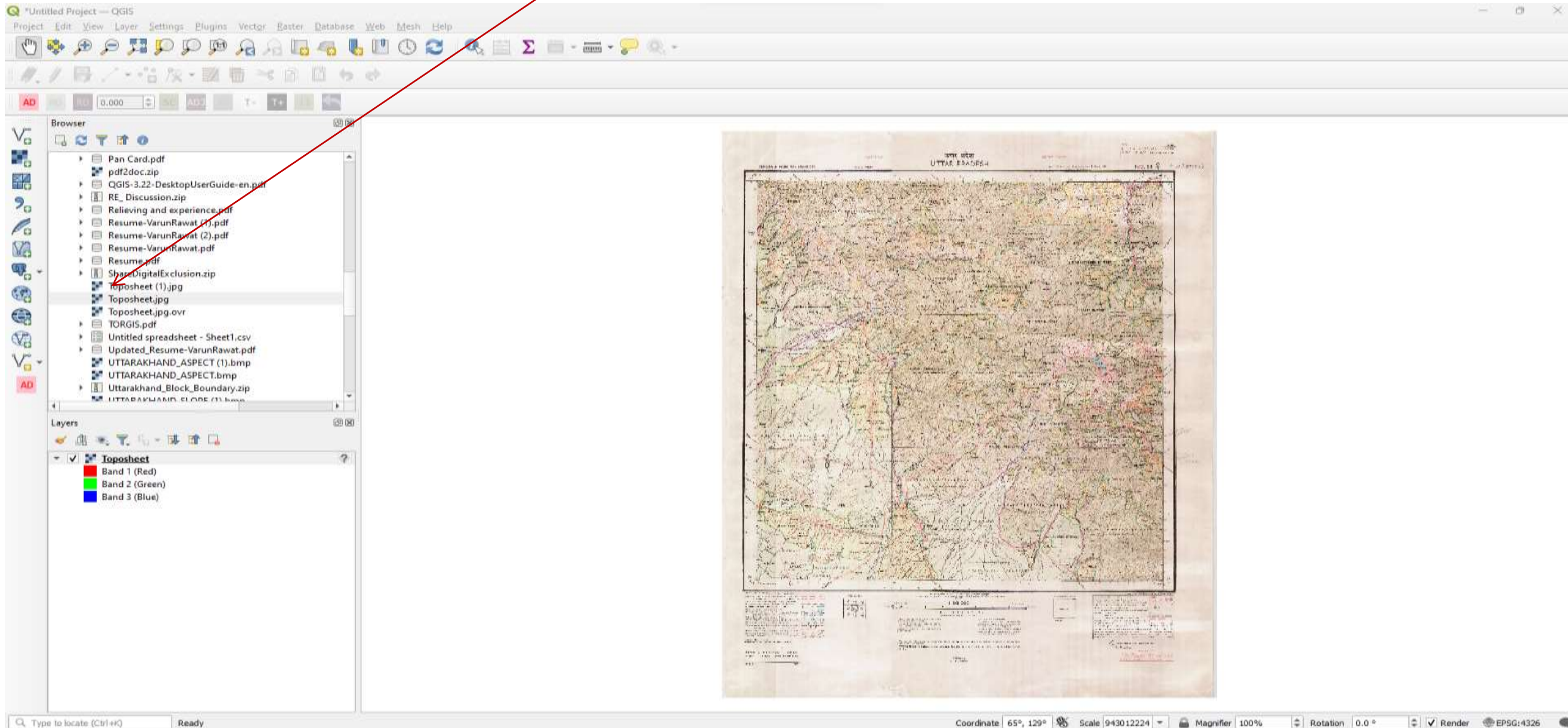
Meaning of Georeferencing ?

Georeferencing ?

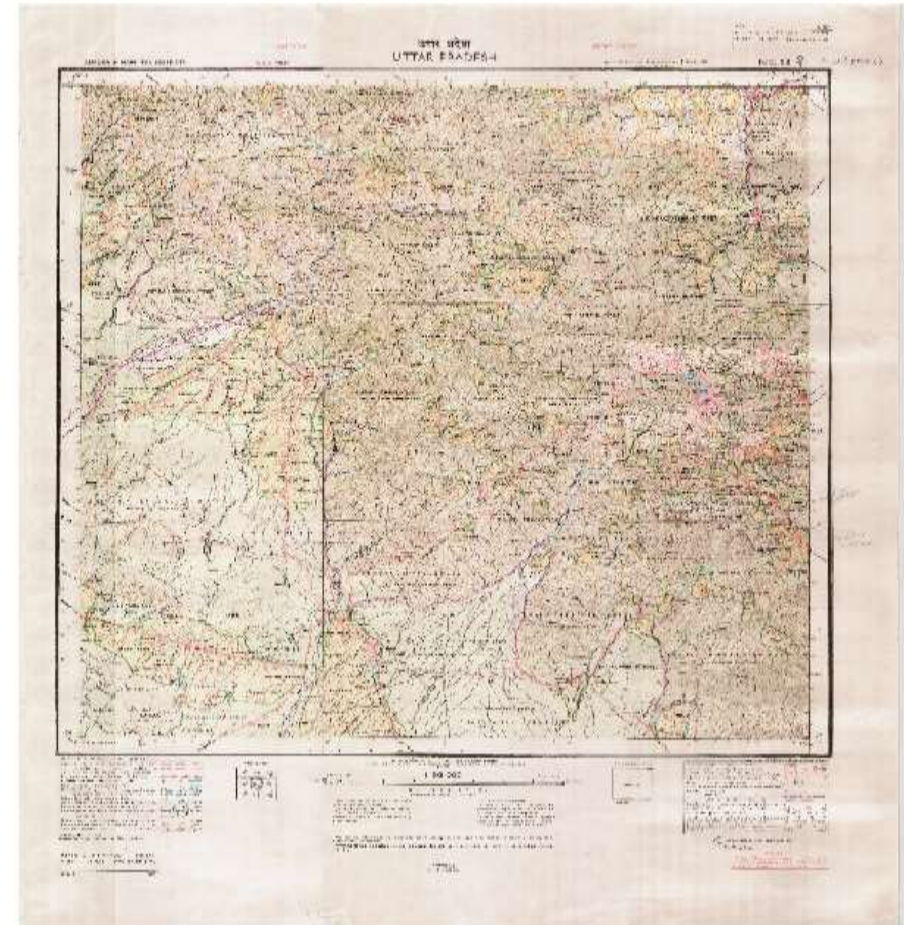
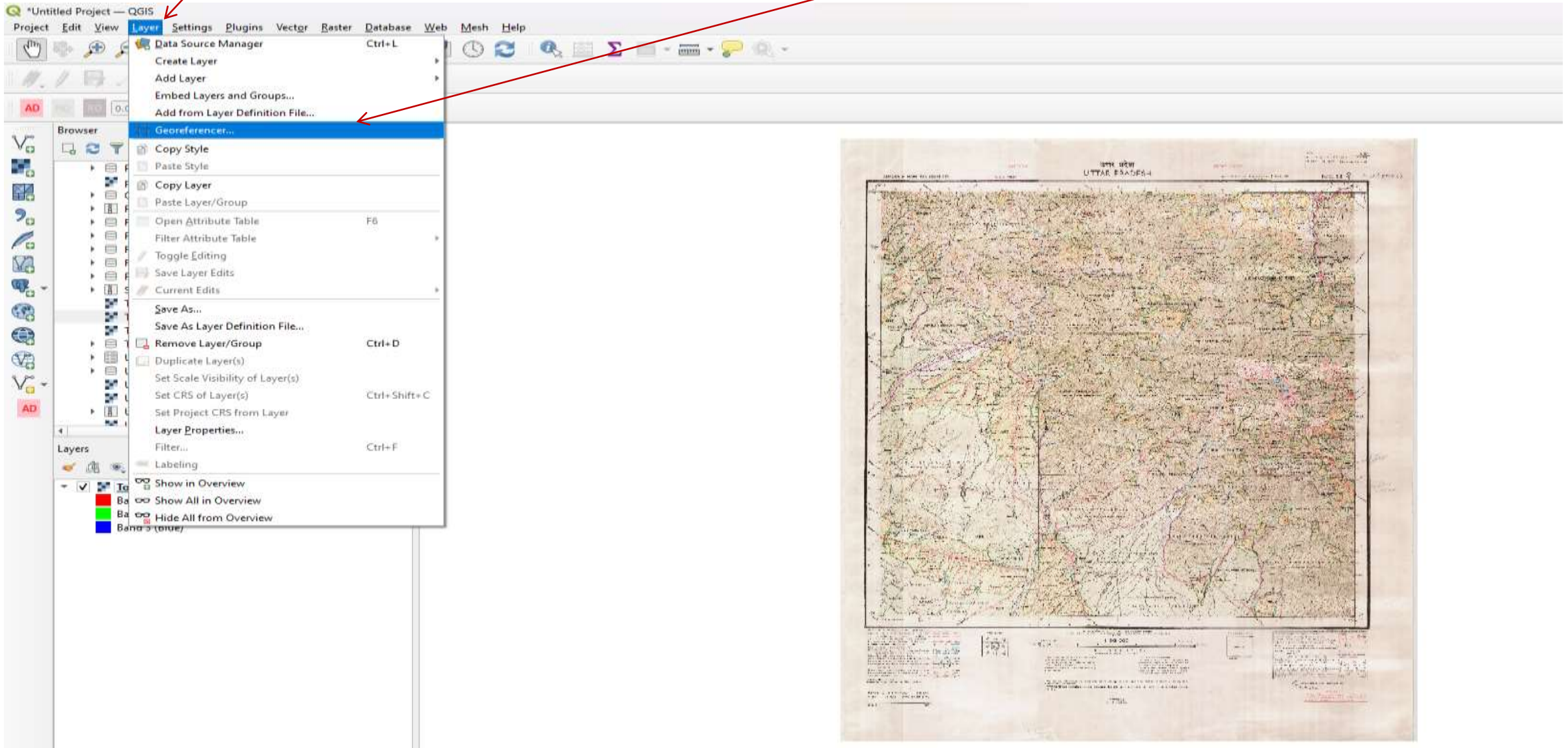
**Where is the
location of the
Area Under
Reference
on
Earths Projection
(Coordinate System) ?**



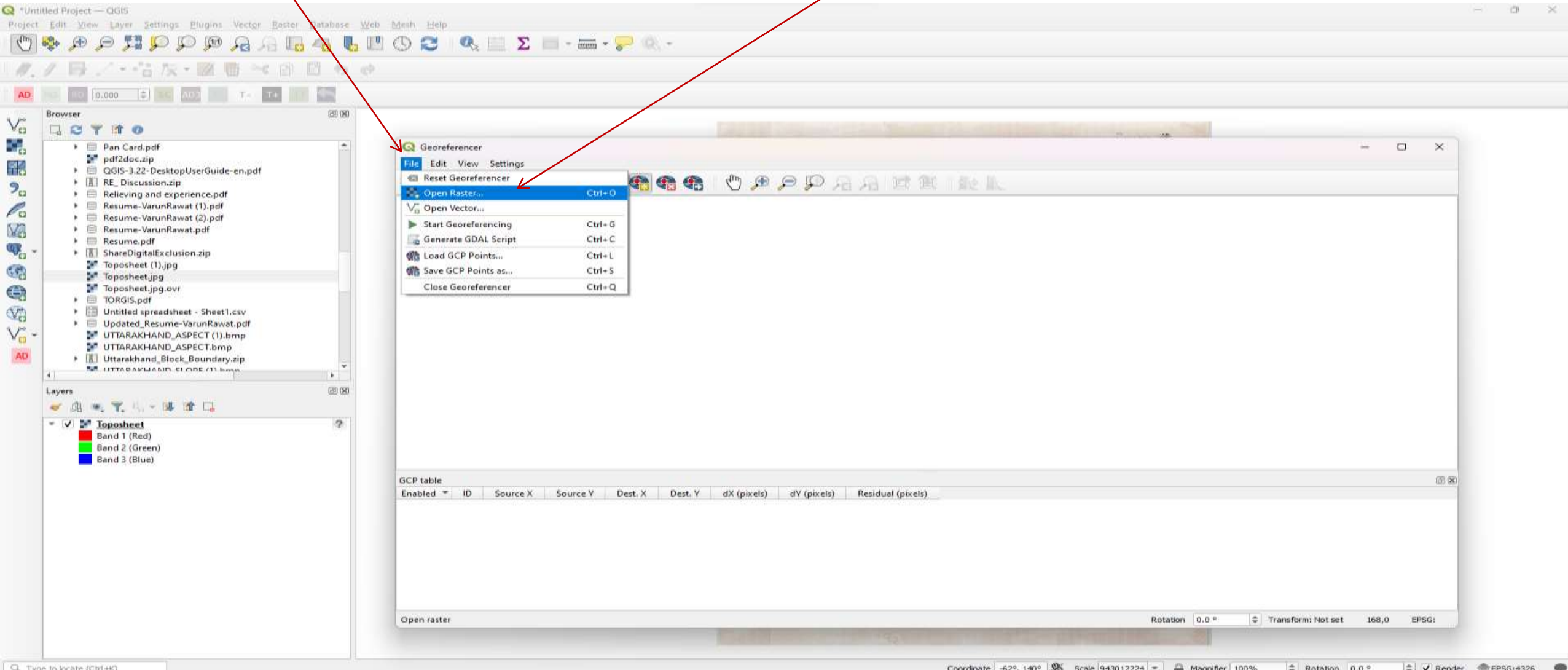
Select Toposheet to be Georeferenced



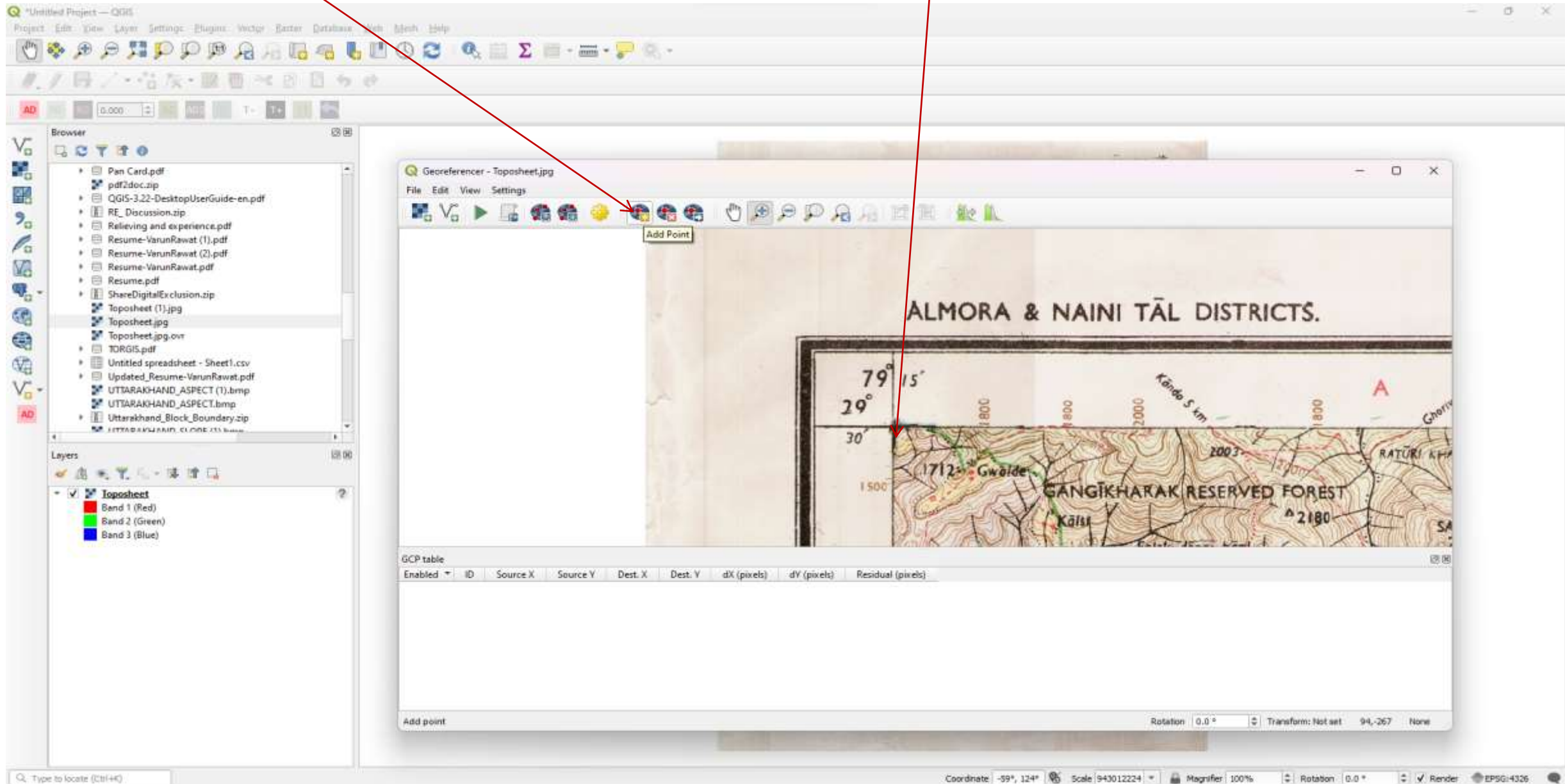
Select Module- 4 and Click on Georeference



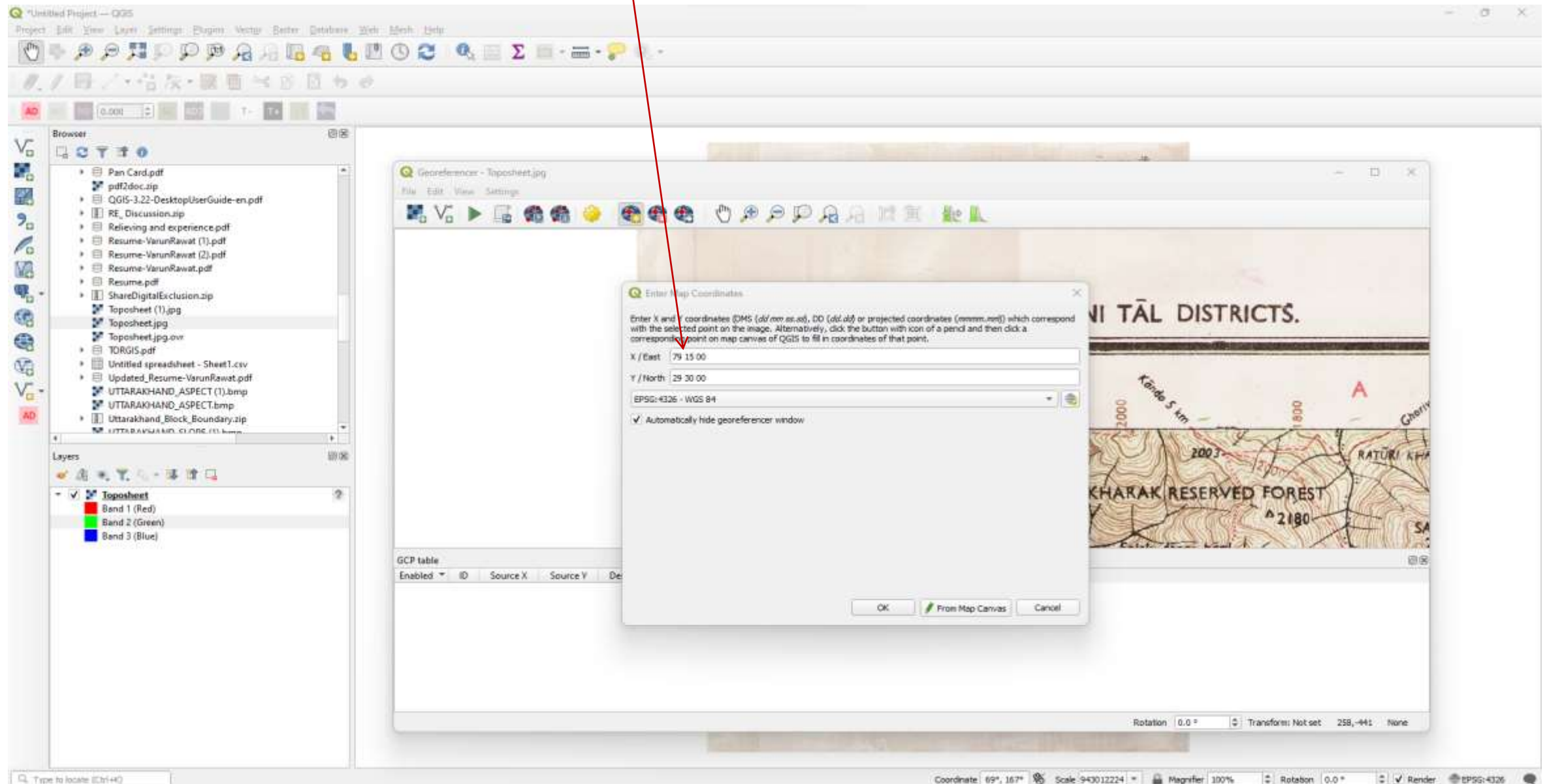
Select File Menu and Click on Open Raster



Select Add Point (First)to be Georeferenced



Enter Coordinates of First Point: 79 15 00 & 29 30 00



Select Second Point to be Georeferenced:

The screenshot shows the QGIS Georeferencer interface. The main window displays a topographic map titled 'No. 53 0/7 R-34(DMMC)' with a coordinate grid. A red arrow points to a point on the map labeled 'Chhota'. The 'Enter Map Coordinates' dialog box is open, showing fields for X/East and Y/North coordinates, and a 'From Map Canvas' button.

Georeferencer - Toposheet.jpg

File Edit View Settings

Enter Map Coordinates

Enter X and Y coordinates (DMS (dd mm ss.ss), DD (dd.dd) or projected coordinates (mmmm.mm)) which correspond with the selected point on the image. Alternatively, click the button with icon of a pencil and then click a corresponding point on map canvas of QGIS to fill in coordinates of that point.

X / East

Y / North

EPSG:4326 - WGS 84

☒ Automatically hide georeferencer window

OK From Map Canvas Cancel

Rotation 0.0° Transform: Linear Translation (79.2288, 2.1111)

Coordinate 30.686°, 81.227° Scale 1:956330 Magnifier 100% Rotation 0.0° Render EPSG:4326

Browser

- Map Service UK Varun
- QGIS 3.32.0
- Shape Files
 - 1 District wise .shp Files
 - UK State level Shape Files
 - Administrative
 - State_Boundary.shp
 - State_Boundary.zip
 - Uttarakhand_Block_Boundaries.shp
 - Uttarakhand_Block_Boundaries.zip
 - Uttarakhand_District_Boundaries.shp
 - Uttarakhand_District_Boundaries.zip
 - Uttarakhand_Tehsil_Boundaries.shp
 - Uttarakhand_Tehsil_Boundaries.zip
 - Aspect
 - Climatic Regions
 - Department of Revenue
 - Department of Urban Development
 - Disaster Management

Layers

- Uttarakhand_Major_Rivers
- State_Boundary
- Uttarakhand_District_Boundary
- ☒ Toposheet_georeferenced
 - Band 1 (Red)
 - Band 2 (Green)
 - Band 3 (Blue)
- OpenStreetMap

GCP table

Enabled	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
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Enter Coordinates of Second Point: 79 30 00 & 29 15 00

QGIS Georeferencer - Toposheet.jpg

Enter Map Coordinates

Enter X and Y coordinates (DMS (dd mm ss.ss), DD (dd.dd) or projected coordinates (xxxxx,yyyy)) which correspond with the selected point on the image. Alternatively, click the button with icon of a pencil and then click a corresponding point on map canvas of QGIS to fill in coordinates of that point.

X / East : 79 30 00

Y / North : 29 15 00

EPSG:4326 - WGS 84

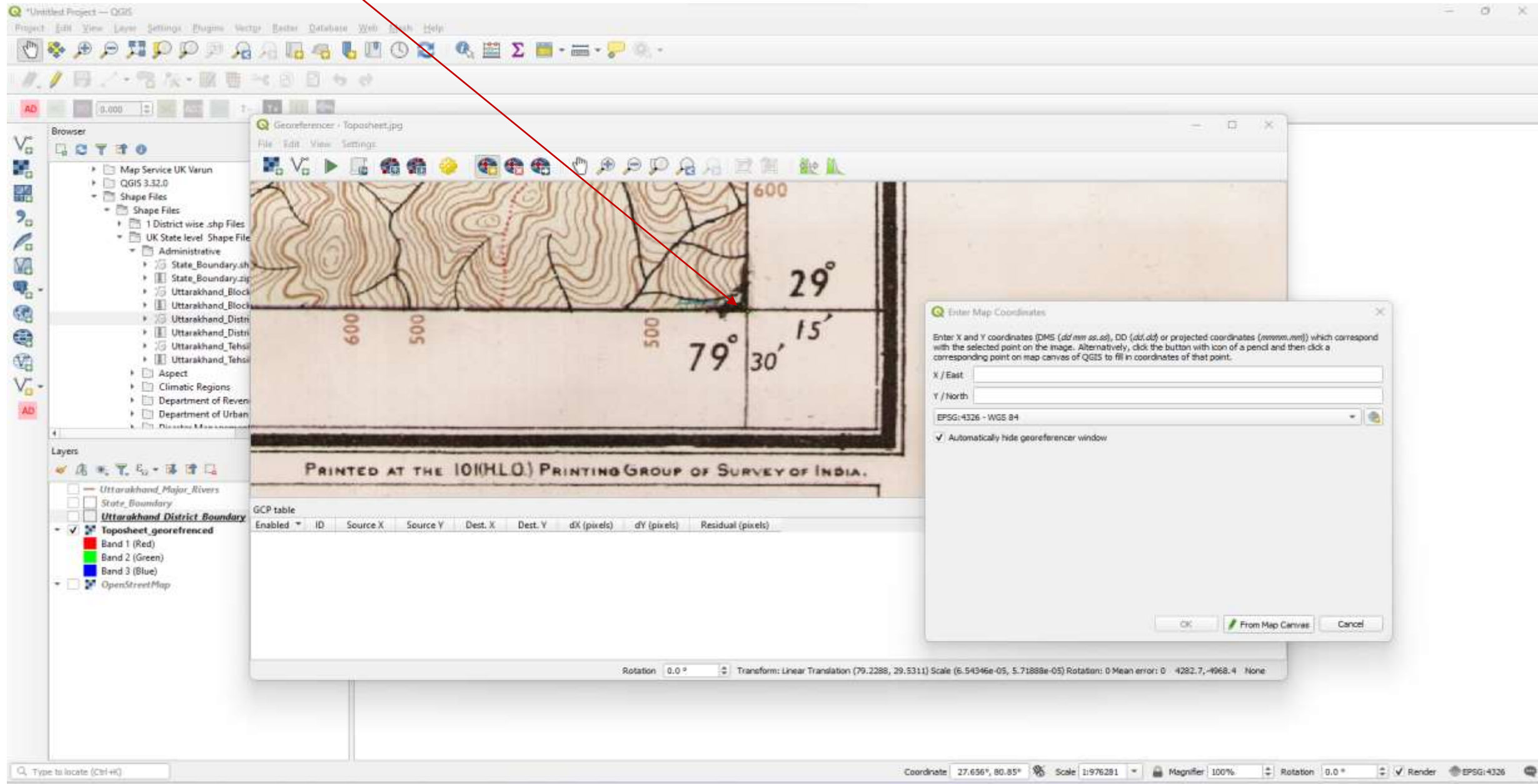
☒ Automatically hide georeferencer window

OK From Map Canvas Cancel

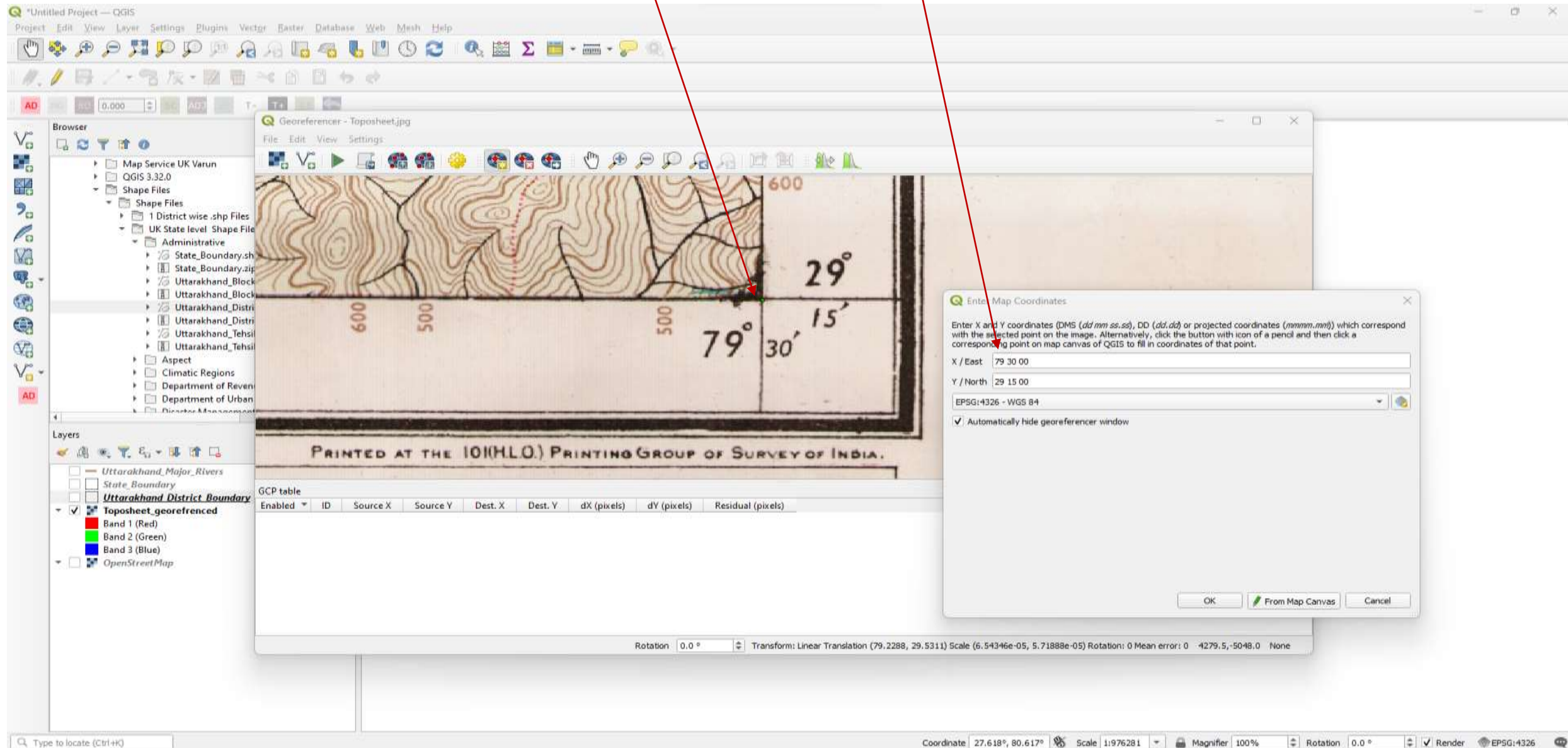
Rotation: 0.0° Transform: Linear Translation (79.2288, 29.5311) Scale (6.54346e-05, 5.71889e-05) Rotation: 0 Mean error: 0 4261, -753 None

Coordinate: 28.938°, 80.055° Scale: 1:956330 Magnifier: 100% Rotation: 0.0° Render EPSG:4326

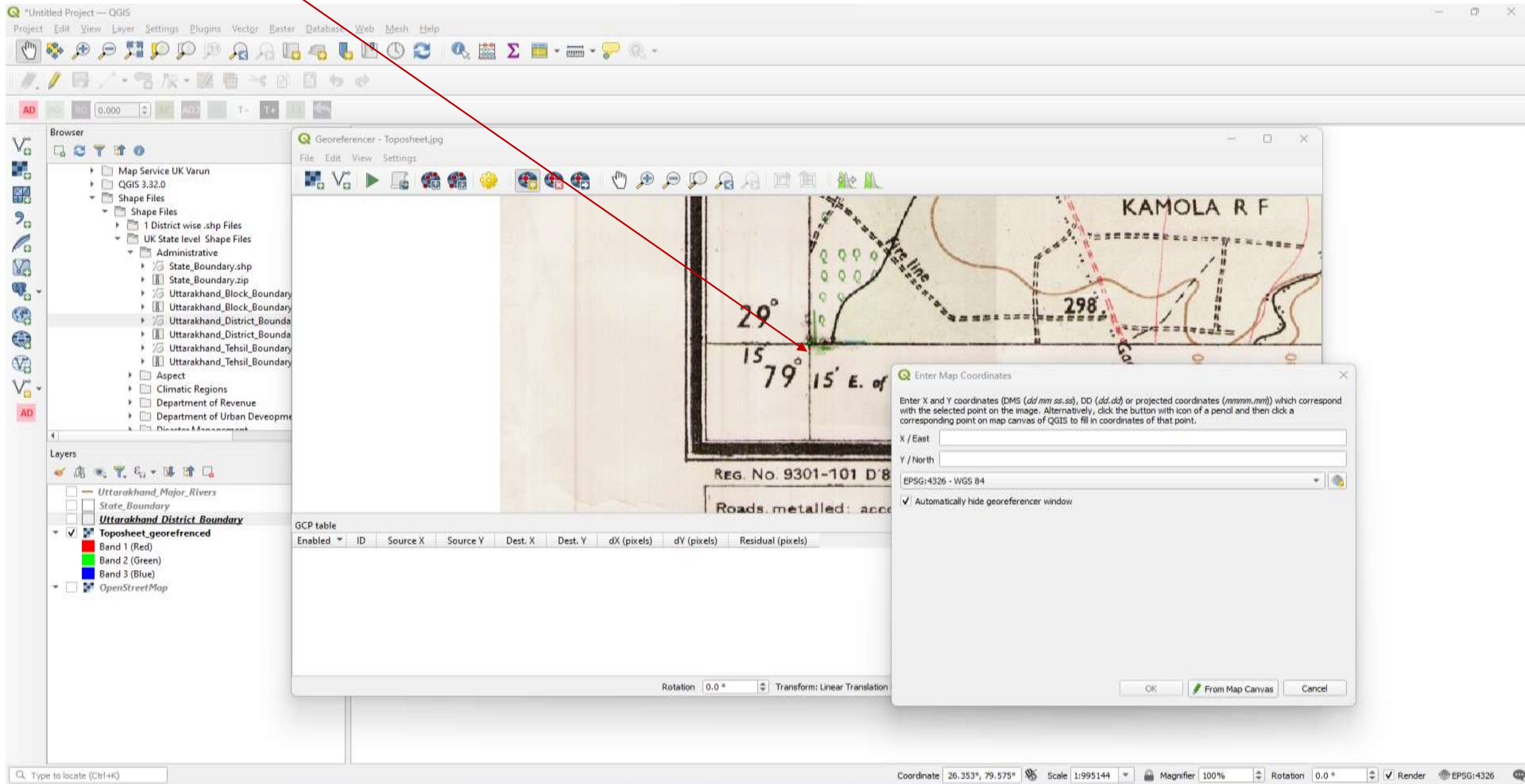
Select Third Point to be Georeferenced:



Enter Coordinates of Third Point: 79 30 00 & 29 15 00



Select Fourth Point to be Georeferenced:



Enter Coordinates of Fourth Point: 79 15 00 & 29 15 00

The screenshot shows the QGIS Georeferencer window with a topographic map of Kamola R F. The map includes labels for '29° 15' 79° 15' E. of', 'REG. No. 9301-101 D 8', and 'Roads metalled: acco'. A dialog box titled 'Enter Map Coordinates' is open, with the following fields:

- X / East: 79 15 00
- Y / North: 29 15 00
- CRS: EPSG:4326 - WGS 84
- ☒ Automatically hide georeferencer window

The dialog box has buttons for 'OK', 'From Map Canvas', and 'Cancel'. The background QGIS window shows a project titled 'Untitled Project - QGIS' with a toolbar and a sidebar containing a 'Browser' panel and a 'Layers' panel. The 'Layers' panel shows a list of layers including 'Uttarakhand Major Rivers', 'State Boundary', 'Uttarakhand District Boundary', 'Toposheet georeferenced', and 'OpenStreetMap'.

GCP table								
Enabled	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)

At the bottom of the QGIS window, the status bar shows the following information:

- Coordinate: 26.36°, 79.651°
- Scale: 1:995144
- Magnifier: 100%
- Rotation: 0.0 °
- Render: ☒
- EPSG:4326

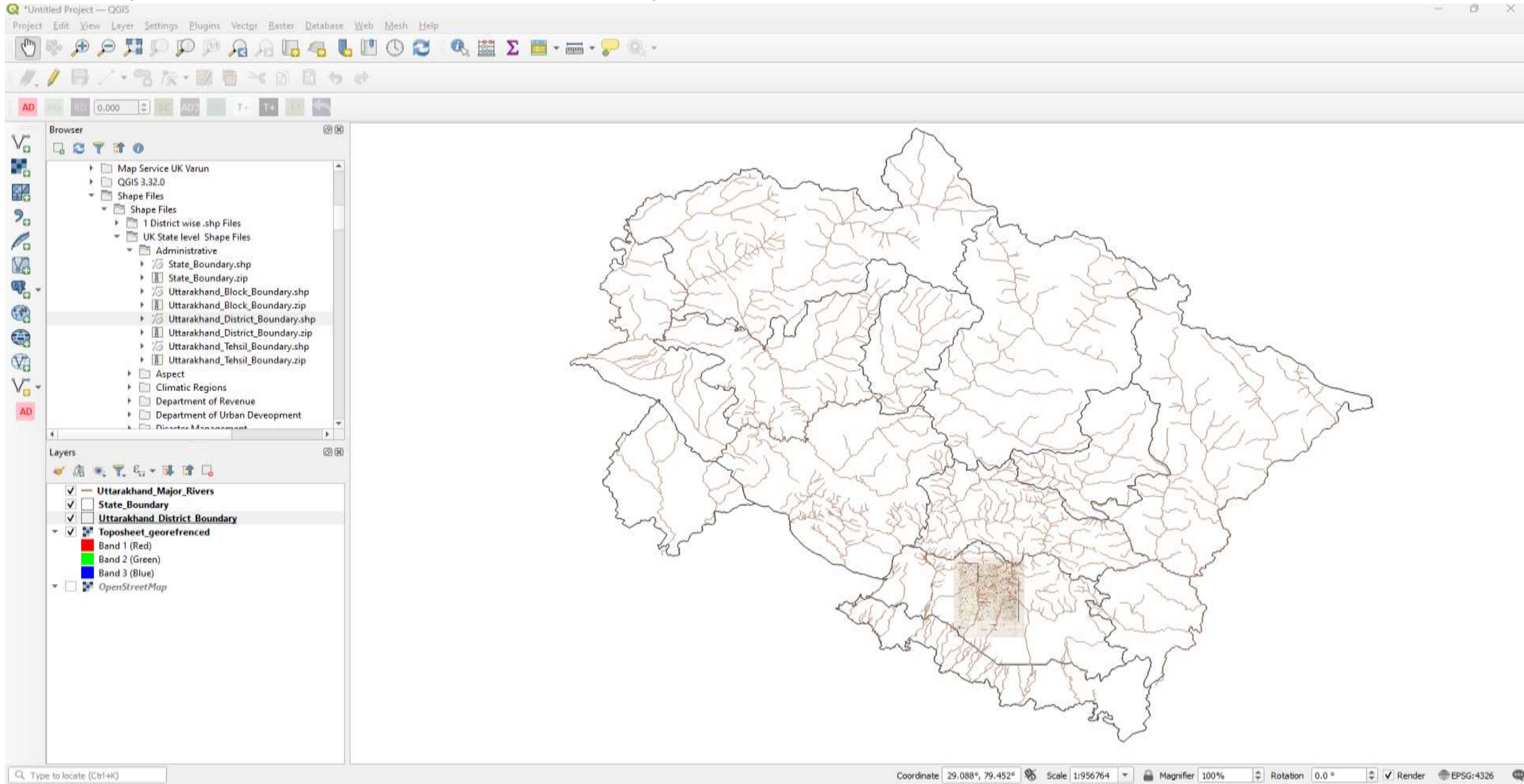
Now all Coordinates of all 4 Points have been Registered

The screenshot displays the QGIS interface with a topographic map loaded. A red arrow points from the text above to the 'GCP table' window. The table lists four Ground Control Points (GCPs) with their source and destination coordinates, pixel offsets, and residuals.

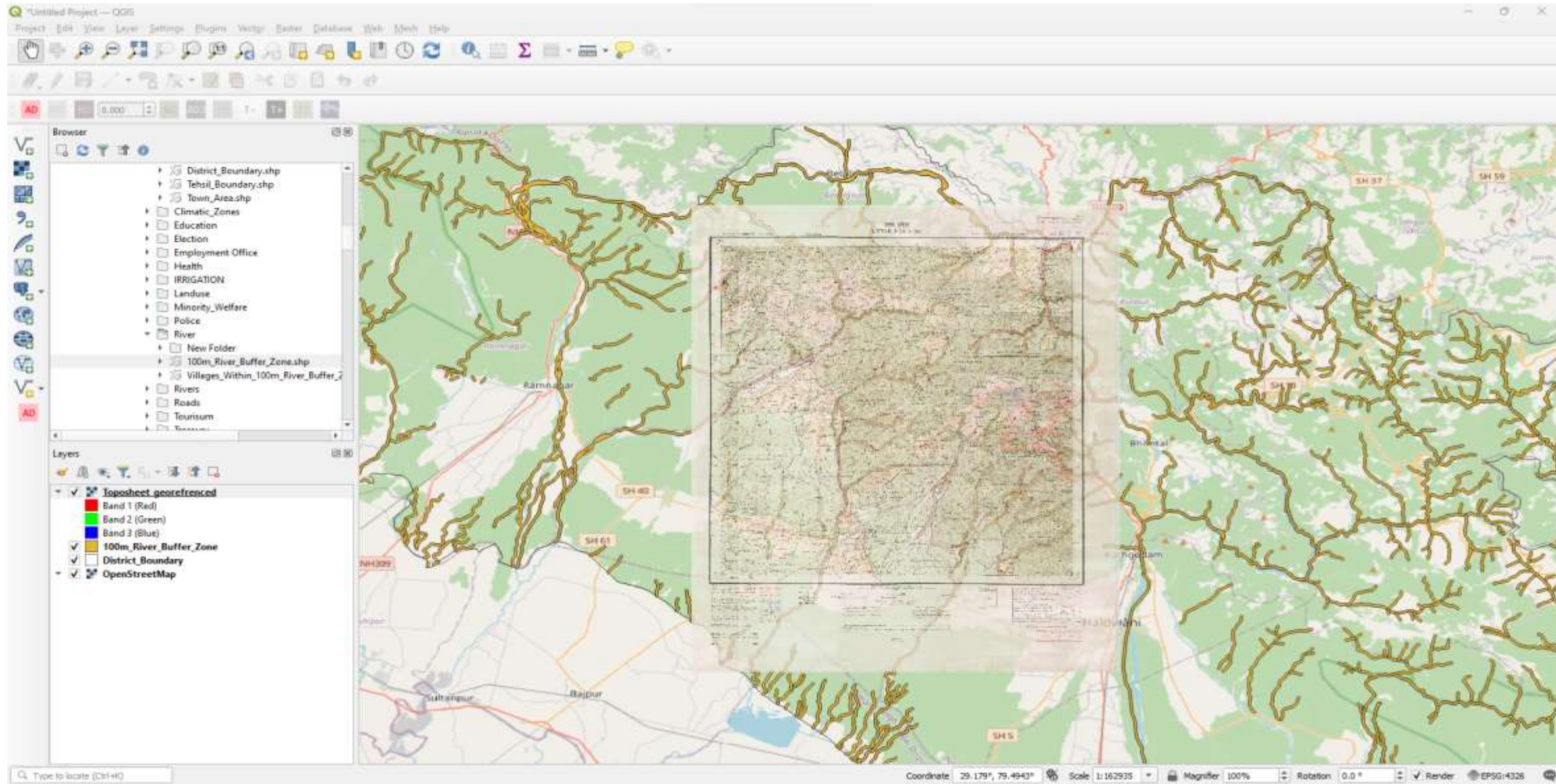
Enabled	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
✓	0	323.229963	-544.126554	79.250000	29.500000	-3.008750	4.208915	5.173736
✓	1	4138.0611	-552.576551	79.500000	29.500000	2.892041	-4.241082	5.133291
✓	2	317.221551	-4901.1488	79.250000	29.250000	2.999861	7.246027	7.842377
✓	3	4143.8361	-4915.6087	79.500000	29.250000	-2.882953	-7.213861	7.768604

At the bottom of the QGIS window, the status bar shows: Rotation: 0.0°, Transform: Linear Translation (79.229, 29.5314) Scale (8.54325e-05, 8.73387e-05) Rotation: 0 Mean error: 9.3834 381/-6012 None EPSG:4326

Open Uttarakhand State Boundary with District Boundaries



Open OSM Map



Thanks...

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